Functional & Performance Testing Template

Model Performance Test

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass /Fail
FT-01	Applicant Details Input Validation	Enter valid and invalid applicant details (Age, Income, Credit Score, Employment, etc.)	Valid inputs are accepted and invalid inputs show validation errors	Application validated inputs correctly	**Pass**
FT-02	Numeric Input Validation	Enter income, debt, age and credit score within and outside valid limits	Accept valid values and reject invalid values	Input validation worked correctly	**Pass**
FT-03	Credit Card Approval Prediction	Enter complete applicant information and click **Predict**	System predicts **Approved** or **Rejected** with confidence score	Prediction displayed successfully	**Pass**
FT-04	Machine Learning Model Loading	Start the Flask application and load the trained model	Model loads successfully without errors	Model loaded successfully	**Pass**

		(.joblib)			
FT-05	Prediction Accuracy Check	Test the application using known dataset records	Prediction matches expected output with high accuracy	Accurate predictions obtained	**Pass**
FT-06	User Interface Validation	Open the web application in different browsers	User interface loads correctly and is responsive	UI displayed correctly	**Pass**
PT-01	Prediction Response Time	Submit applicant details and measure prediction time	Prediction should complete within **2 seconds**	Average response time: **1.2 seconds**	**Pass**
PT-02	Multiple Prediction Requests	Submit multiple prediction requests consecutively	Application remains responsive without slowdown	Stable performance observed	**Pass**
PT-03	Load Test	Simulate multiple users accessing the application simultaneously	System should continue functioning without crashes	Application remained stable	**Pass**
PT-04	Model Performance Test	Evaluate Logistic Regression, Random Forest and Decision Tree models	Best-performing model should be selected based on accuracy	Random Forest achieved highest accuracy and was selected	**Pass**

Test Summary

Category	Total	Passed	Failed
Functional Tests	6	6	0
Performance Tests	4	4	0
Total	**10**	**10**	**0**

Conclusion

The **Credit Card Approval Prediction** system successfully passed all functional and performance tests. The application correctly validates user inputs, loads the trained machine learning model, predicts credit card approval status accurately, provides fast response times, and remains stable under multiple requests. The system satisfies the functional and performance requirements and is ready for deployment.